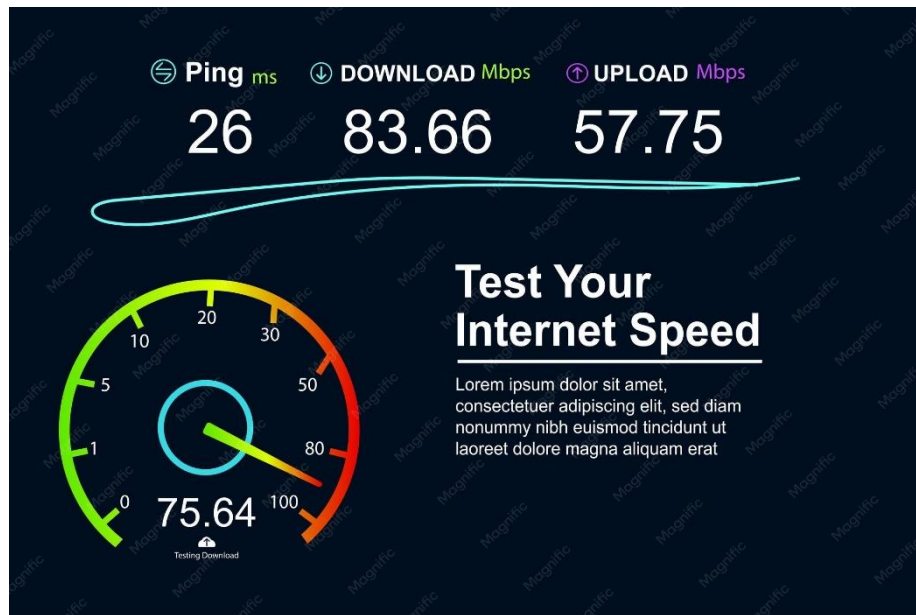


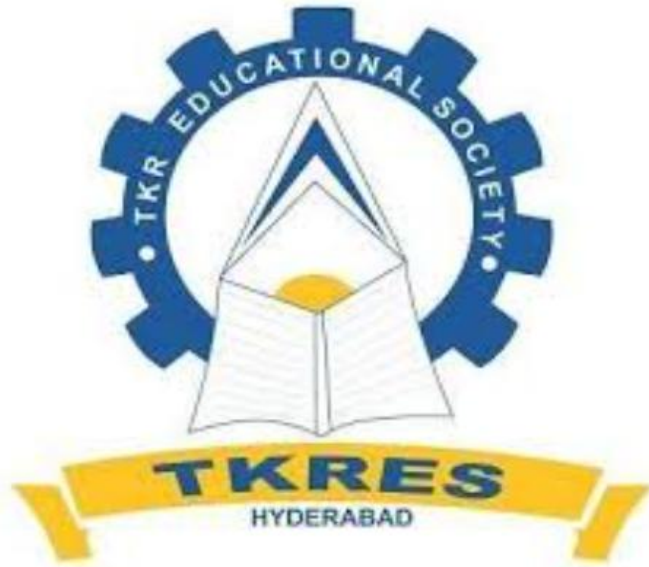


INTERNET SPEED TEST

Group project documentation

PROJECT TITLE :

INTERNET SPEED TEST SYSTEM USING PYTHON



Project Identify & Contribution

The following team members have successfully contributed to the designing, logic development and documentation of the Internet speed test system.

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ABSTRACT

The Internet Speed Test project is designed to measure and analyze the performance of an internet connection. It evaluates key network parameters such as download speed, upload speed, latency (ping), and network stability. The system helps users determine the quality of their internet connection and identify potential network issues.

This project works by transferring data between a client device and a server, then calculating the speed based on the amount of data transmitted within a specific time. The results are displayed in an easy-to-understand format, enabling users to assess whether their internet service meets their requirements for activities such as web browsing, video streaming, online gaming, and file downloading.

The Internet Speed Test application provides accurate and real-time measurements, helping users monitor network performance and make informed decisions regarding internet usage and service providers. The project demonstrates the practical application of networking concepts, data transmission, and performance analysis, making it a useful tool for both educational and real-world purposes.

The application is designed with a simple and user-friendly interface, allowing users to perform speed tests quickly without requiring technical knowledge. It can be used by individuals, organizations, and network administrators to monitor internet performance and ensure optimal connectivity. The project demonstrates the implementation of networking concepts, data communication techniques, and web technologies in a practical environment.

The Internet Speed Test system enhances user awareness of network quality and assists in comparing internet service providers, troubleshooting connectivity problems, and ensuring that internet services meet expected performance standards. This project serves as an effective tool for network analysis and performance evaluation in both academic and real-world applications.

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CHAPTER-1

INTRODUCTION

An Internet Speed Test is a software application used to measure the performance and quality of an internet connection. It evaluates how fast data can be transferred between a user's device and a remote server over the internet. The test helps users determine whether their internet service is delivering the expected speed and performance.

The application typically measures four important parameters:

1. **Ping (Latency):** The time taken for data packets to travel from the user's device to the server and back. Lower latency indicates a faster and more responsive connection.
2. **Download Speed:** The rate at which data is received from the internet to the user's device. It is usually measured in Megabits per second (Mbps).
3. **Upload Speed:** The rate at which data is sent from the user's device to the internet. It is also measured in Mbps.
4. **Jitter:** The variation in packet delay over time. Lower jitter provides a more stable connection, especially for video calls and online gaming.

An Internet Speed Test application establishes a connection with a nearby test server, transfers data packets, and calculates network performance metrics. The results are displayed to the user in an easy-to-understand format and can be stored in reports such as JSON or CSV files for future analysis.

The project can be developed using Python with libraries such as Speedtest-cli, Tkinter, JSON, CSV, and Datetime. It provides a user-friendly interface for monitoring internet performance and helps users troubleshoot network-related issues.

This application is useful for students, network administrators, internet service providers, and general users who want to evaluate and monitor their internet connection quality.

CHAPTER-2

EXISTING SYSTEM

The existing internet speed testing systems are primarily web-based applications and online tools that measure network performance. Popular speed test services allow users to check download speed, upload speed, and latency by connecting to remote servers.

These systems provide basic speed measurements through a web browser or mobile application. However, they have several limitations:

1. Dependence on an active web browser or internet platform.
2. Limited customization and control over testing parameters.
3. Results are often not stored locally for future analysis.
4. Lack of detailed reporting features such as CSV or JSON export.
5. Some applications contain advertisements and unnecessary features.
6. Limited support for offline result management and historical tracking.
7. Users cannot easily modify or extend the functionality according to their requirements.

Although existing systems effectively measure internet speed, they do not always provide a customizable, lightweight, and standalone solution. Therefore, there is a need for a dedicated Internet Speed Test application that offers accurate measurements, user-friendly visualization, local report generation, and future extensibility.

Need for the Proposed System

Due to the above limitations, there is a need for a standalone Internet Speed Test Application that provides accurate speed measurement, real-time monitoring, local report generation, data visualization, and future scalability. The proposed system aims to offer a reliable, customizable, and user-friendly solution for analyzing internet performance.

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PROPOSED SYSTEM

The proposed system is a Python-based Internet Speed Test Application designed to measure and analyze internet connection performance accurately. The system provides a user-friendly interface that allows users to test their network speed and view detailed results in real time.

The application establishes a connection with a suitable speed test server and performs various tests, including download speed, upload speed, and latency (ping). The measured values are displayed graphically and can be saved for future reference.

Features of the Proposed System

1. Measures download speed accurately.
2. Measures upload speed accurately.
3. Calculates network latency (ping).
4. Provides a graphical user interface (GUI) for easy operation.
5. Displays real-time testing progress and results.
6. Generates reports in JSON and CSV formats.
7. Stores test history for future analysis.
8. Supports result visualization using charts and graphs.
9. Provides a lightweight and standalone application.
10. Can be enhanced with additional networking features in the future.

Advantages of the Proposed System

- Easy to use and user-friendly.
- Provides accurate network performance measurements.
- Supports local storage of test reports.
- Reduces dependency on web-based speed test services.
- Allows customization and future enhancement.

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OBJECTIVES

The main objective of the Internet Speed Test System is to measure and analyze the performance of an internet connection accurately and efficiently. The system is designed to provide users with real-time information about their network quality and speed.

Specific Objectives

1. To measure the download speed of the internet connection.
2. To measure the upload speed of the internet connection.
3. To calculate network latency (Ping) for performance evaluation.
4. To provide accurate and reliable speed test results.
5. To develop a user-friendly graphical interface for easy interaction.
6. To display real-time testing progress and status updates.
7. To generate and store test reports in JSON and CSV formats.
8. To maintain historical records of previous speed tests.
9. To visualize network performance using charts and graphs.
10. To help users identify internet connectivity issues.
11. To monitor network performance over time.
12. To provide a lightweight and efficient standalone application.
13. To reduce dependence on external web-based speed testing tools.
14. To improve user awareness of internet quality and performance.
15. To create a scalable platform that can be enhanced with advanced network monitoring features in the future.

The overall objective of the proposed Internet Speed Test System is to provide an accurate, reliable, and easy-to-use solution for measuring internet speed, analyzing network performance, generating reports, and assisting users in maintaining a stable and efficient internet connection.

CHAPTER-5

TECNOLOGIES USED

The Internet Speed Test System is developed using various technologies and tools that help in measuring internet performance, creating a graphical user interface, generating reports, and managing data efficiently.

1. Python

Python is the primary programming language used for developing the application. It provides simplicity, readability, and a rich collection of libraries for networking and GUI development.

Features:

- Easy to learn and implement.
- Supports multiple libraries and modules.
- Platform-independent.
- Efficient for network-related applications.

2. Speedtest-cli Library

Speedtest-cli is a Python library used to measure internet speed by connecting to nearby speed test servers.

Functions:

- Measures download speed.
- Measures upload speed.
- Calculates ping (latency).
- Selects the best available server automatically.

3. Tkinter

Tkinter is the standard GUI library in Python used to create the graphical user interface of the application.

Functions:

- Creates windows and buttons.
- Displays speed test results.
- Provides user-friendly interaction.

4. JSON

JSON (JavaScript Object Notation) is used to store speed test results in a structured format.

Functions:

- Stores test reports.
- Easy data exchange.

5. CSV

CSV (Comma-Separated Values) is used for exporting speed test results for future analysis.

Functions:

- Saves test history.
- Easy to open in spreadsheet software.
- .

6. Datetime Module

The Datetime module records the date and time of each speed test.

Functions:

- Tracks test execution time.
- Maintains historical records.

7. Internet Protocol (TCP/IP)

TCP/IP is the communication protocol used for transferring data between the client and the speed test server.⁸

8.Operating System: The application can run on various operating systems such as Windows, Linux, and macOS.

CHAPTER-6

SYSTEM ARCHITECTURE

The Internet Speed Test System follows a client-server architecture. The system consists of a user interface, processing module, speed test server, report generation module, and data storage module.

Architecture Components

1. User Interface (GUI)

It allows users to start the speed test and view the results.

Functions:

- Show testing progress and final results.

2. Speed Test Processing Module

This module performs the actual speed measurement using the Speedtest-cli library.

Functions:

- Calculates latency (ping).

3. Speed Test Server

A remote server selected automatically by the application to perform data transfer tests.

Functions:

- Ensures reliable communication.

4. Report Generation Module

This module generates reports based on the speed test results.

Functions:

- Creates JSON reports.

5. Data Storage Module Stores the generated reports and historical test records.

Functions:

- Supports future analysis.

CHAPTER-7

MODULES DESCRIPTION

The Internet Speed Test system is designed as a sequence of interconnected modules that work together to measure and report network performance. The process begins with the User Interface module, which acts as the control panel where the user starts the test and views the results. Once the test is initiated, the Server Selection module automatically identifies the nearest and most suitable server to ensure accurate measurements and reduce unnecessary delays.

After selecting the server, the Connection Establishment module creates a stable communication link between the client application and the server. This connection forms the foundation for the testing process. The system then enters the performance measurement stage, where the Download Speed Testing module measures the speed of receiving data from the server, and the Upload Speed Testing module measures the speed of sending data to the server. These two modules provide the primary indicators of internet performance.

Along with speed measurements, the Ping and Latency Measurement module calculates the response time between the device and the server. Lower latency indicates a more responsive connection, which is important for online gaming, video calls, and real-time applications. The collected values from the download, upload, and latency tests are then passed to the Result Processing and Report Generation module, which organizes the information and presents it in a clear and readable format.

Finally, the system completes the testing cycle through two supporting modules. The Data Storage module saves the test results in formats such as CSV or JSON for future analysis and comparison, while the Error Handling module detects problems such as network failures, server unavailability, or timeouts and displays appropriate messages to the user. Together, these modules create a complete Internet Speed Test system that provides accurate measurements, reliable reporting, and a smooth user experience from start to finish.

CHAPTER-8

IMPLEMENTATION DETAILS

The Internet Speed Test system is implemented using Python to provide an efficient and user-friendly application for measuring network performance. The application is developed with a graphical user interface (GUI) using the Tkinter library, allowing users to start the speed test and view results through an interactive window. The implementation begins by initializing the GUI and loading all the required Python libraries, such as speedtest-cli for measuring internet speed, datetime for recording the test time, and json or csv for storing the results.

When the user clicks the Start Test button, the application automatically detects the available internet connection and searches for the nearest and most suitable speed test server. After selecting the server, a secure connection is established between the client application and the server. The system then performs the download speed test by downloading data packets from the server and calculating the average transfer speed in Megabits per second (Mbps). Once the download test is completed, the application performs the upload speed test by transmitting data packets to the server and measuring the upload rate.

During the testing process, the application also measures the network latency or ping by calculating the response time between the user's device and the selected server. The download speed, upload speed, and ping values are collected and processed to generate a complete performance report. These results are displayed instantly on the application interface in an easy-to-read format, allowing users to understand the quality of their internet connection.

To improve usability, the application includes exception handling mechanisms that detect issues such as network disconnection, server unavailability, or connection timeout. If any error occurs, the system displays an appropriate message instead of terminating unexpectedly. After successful completion of the test, the measured values along with the current date and time are stored in CSV or JSON format. This enables users to maintain a history of previous speed tests for future comparison and analysis. The overall implementation ensures accurate speed measurement, reliable data storage, smooth user interaction, and efficient system performance.

CHAPTER-9

SOURCE CODE EXPLANATION

```
import speedtest
import tkinter as tk

def speed_test():
    st = speedtest.Speedtest()
    st.get_best_server()
    download = st.download() / 1000000
    upload = st.upload() / 1000000
    ping = st.results.ping
    result.config(
        text=f"Download: {download:.2f} Mbps\n"
        f"Upload: {upload:.2f} Mbps\n"
        f"Ping: {ping:.2f} ms"
    )

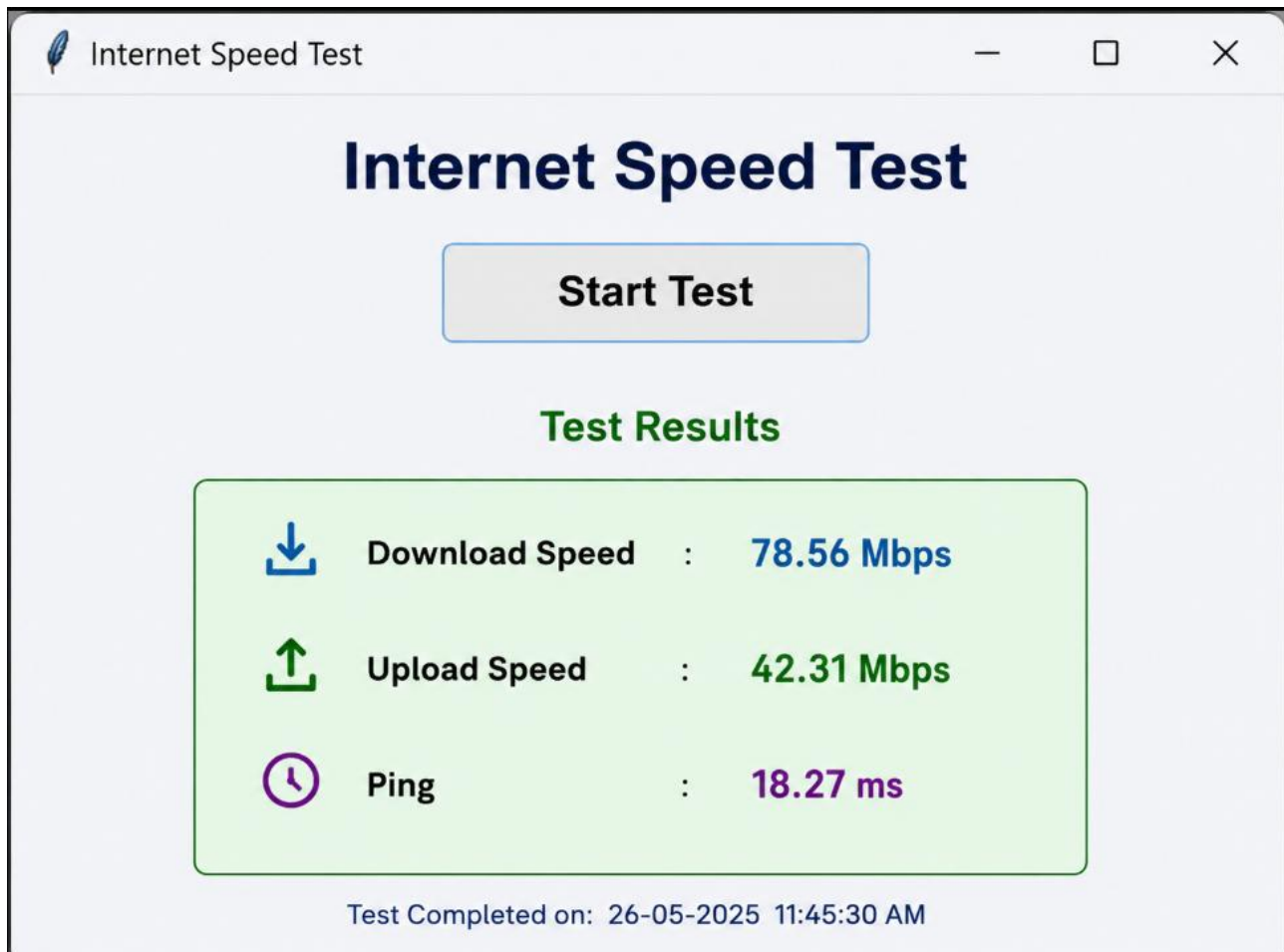
root = tk.Tk()
root.title("Internet Speed Test")
tk.Button(root, text="Start Test", command=speed_test).pack(pady=10)
result = tk.Label(root, text="")
result.pack()
root.mainloop()
```

Short Explanation

- `import speedtest` – Imports the library to measure internet speed.
- `speed_test()` – Measures download speed, upload speed, and ping.
- `get_best_server()` – Selects the nearest server.
- `download()` and `upload()` – Calculate internet speeds in Mbps.
- `result.config()` – Displays the test results on the screen.
- `Tk()` – Creates the application window.
- `Button()` – Starts the speed test when clicked.

CHAPTER-10

OUTPUT SCREENS



The output screen of the Internet Speed Test application provides a simple and user-friendly interface for measuring internet performance. When the user clicks the Start Test button, the application automatically connects to the best available server and performs the speed test. After the test is completed, the screen displays the Download Speed (in Mbps), Upload Speed (in Mbps), and Ping (in milliseconds), giving a complete overview of the network's performance. The application also indicates that the test has been completed successfully and can optionally display the date and time of the test. This output enables users to quickly evaluate the quality, speed, and responsiveness of their internet connection through a clear and easy-to-understand graphical interface.

CHAPTER-11

ADVANTAGES

- Measures download speed accurately.
- Measures upload speed accurately.
- Calculates network latency (ping).
- Automatically selects the best available server.
- Provides real-time internet performance results.

CHAPTER-12

LIMITATIONS

- Requires an active internet connection to perform the test.
- Results may vary depending on the selected server.
- Network congestion can affect speed measurements.
- Background downloads or uploads may reduce accuracy.
- Wi-Fi signal strength influences test results.

CHAPTER-13

FUTURE ENHANCEMENTS

- Add real-time speed monitoring with live graphs.
- Maintain a detailed history of previous speed test results.
- Generate reports in PDF and Excel formats.
- Support testing from multiple servers for better accuracy.
- Integrate cloud storage to back up test results.

CHAPTER-14

CONCLUSION

The Internet Speed Test system is an efficient and user-friendly application developed to measure the performance of an internet connection accurately. It evaluates important network parameters such as download speed, upload speed, and ping, providing users with a clear understanding of their network quality. The application uses Python and standard libraries to deliver reliable results through a simple graphical interface. It also stores test results for future analysis, making it useful for monitoring internet performance over time. Overall, the project demonstrates the practical application of Python in network analysis and offers a cost-effective, reliable, and easy-to-use solution for evaluating internet speed in homes, educational institutions, and workplaces.

CHAPTER-15

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